Problem 1

In the charge's rest frame there is only Coulomb electric field

$$\vec{E}' = \frac{1}{4\pi\varepsilon_0}\frac{q\vec{R}'}{R'^3}, \quad \vec{B}' = 0$$

To obtain $\vec{E}(\vec{R})$ and $\vec{B}(\vec{R})$ in the K-frame we need

a) to express $\vec{E}/\vec{B}$ via $\vec{E}'/\vec{B}'$ and

b) to express $\vec{R}'$ via $\vec{R}$.

$$\vec{R} = (x - vt, y, x)$$

b) first:

$$x' = \gamma(x - vt) \quad y' = y \quad x' = x \Rightarrow \boxed{\vec{R}' = \vec{R} + (\gamma - 1)\vec{n}(\vec{n}\cdot\vec{R})}$$

note that $R'^2 = R^2 + (\gamma-1)^2(\vec{n}\cdot\vec{R})^2 = R^2 + \beta^2\gamma^2(\vec{n}\cdot\vec{R})^2$

+2 (γ-1)

a) Using the solution to problem 1 we find

$$\vec{E} = \gamma\vec{E}' - (\gamma - 1)\vec{n}(\vec{n}\cdot\vec{E}')$$

$$\vec{B} = \gamma\frac{\vec{V}}{c^2}\times\vec{E}'$$

Magnetic field is thus:

$$\vec{B} = \frac{\gamma q}{4\pi\varepsilon_0 c^2}\frac{\vec{V}\times\vec{R}}{\left[R^2 + \beta^2\gamma^2(\vec{n}\cdot\vec{R})^2\right]^{3/2}}$$

$$\boxed{\vec{B} = \frac{q\mu_0}{4\pi}\frac{\gamma\vec{V}\times\vec{R}}{\left[R^2 + \beta^2\gamma^2(\vec{n}\cdot\vec{R})^2\right]^{3/2}}}$$

Electric field

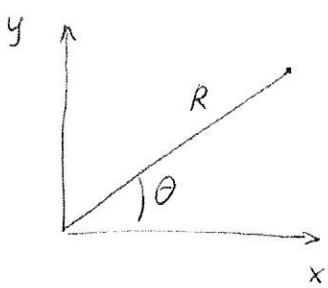
$$\vec{E} = \frac{q}{4\pi\varepsilon_0} \frac{\gamma \vec{R}' - (\gamma-1)(\vec{R}'\cdot\vec{n})\vec{n}}{R'^3} =$$

$$= \frac{q}{4\pi\varepsilon_0} \frac{\gamma\vec{R} + \gamma(\gamma-1)\vec{n}(\vec{n}\cdot\vec{R}) - (\gamma-1)\vec{n}(\vec{n}\cdot\vec{R}) - (\gamma-1)^2\vec{n}(\vec{n}\cdot\vec{R})}{\left[R^2+\beta^2\gamma^2(\vec{n}\cdot\vec{R})^2\right]^{3/2}}$$

$$\boxed{\vec{E} = \frac{q}{4\pi\varepsilon_0} \frac{\gamma\vec{R}}{\left[R^2+\beta^2\gamma^2(\vec{n}\cdot\vec{R})^2\right]^{3/2}}}$$

Note that $\vec{B} = \frac{\vec{V}\times\vec{E}}{c^2}$

Electric field can now be written in more convenient form

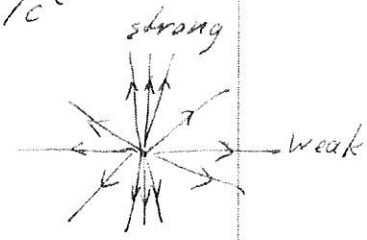


$\vec{n}\cdot\vec{R} = R\cos\theta$

$$R^2+\beta^2\gamma^2(\vec{n}\cdot\vec{R})^2 = R^2 + \frac{V^2/c^2}{1-V^2/c^2}R^2\cos^2\theta$$

$$= R^2 \frac{1-\frac{V^2}{c^2}\sin^2\theta}{1-V^2/c^2}$$

$$\vec{E} = \underbrace{\vec{E}_0(\vec{R})}_{=\frac{q\vec{R}}{4\pi\varepsilon_0 R^3}} \frac{1-\frac{V^2}{c^2}}{\left(1-\frac{V^2}{c^2}\sin^2\theta\right)^{3/2}}$$

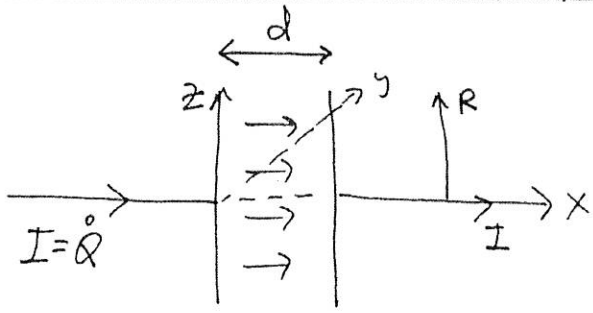


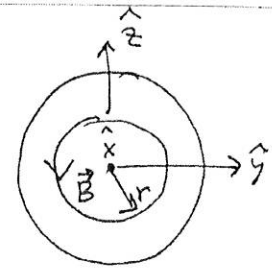
field of a stationary charge. For $\frac{V}{c} \approx 1$ the field is strongly <u>suppressed</u> in longitudinal direction ($\theta = 0$) and <u>enhanced</u> in the transverse direction ($\theta = \frac{\pi}{2}$,

HW 9

P1

a)





Inside capacitor, $\vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{\mu_0 \dot{Q}}{\pi R^2} \hat{x}$ (displacement current)

$$\vec{E} = \frac{\sigma}{\varepsilon_0} \hat{x} = \frac{Q}{\varepsilon_0 \pi R^2} \hat{x}$$

For $r < R$: $\quad 2\pi r B(r) = \mu_0 \frac{\dot{Q}}{\pi R^2} \pi r^2$

$$\vec{B}(r) = \frac{\mu_0 \dot{Q} r}{2\pi R^2} \hat{\varphi}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = 0 \quad \text{because } \dot{Q} = \text{const}$$

$\Rightarrow \vec{B}$ does not generate additional el. field.

$$\begin{cases} \vec{E} = \dfrac{Q}{\varepsilon_0 \pi R^2} \hat{x} \\ \vec{B} = \dfrac{\mu_0 \dot{Q}}{2\pi R^2} \left(y \hat{z} - z \hat{y} \right) \end{cases}$$

$\leftarrow$ note that (y, z) is the plane of capacitor's plate on these notations.

Pointing $\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B} = \frac{Q \dot{Q}}{2\pi^2 R^4 \varepsilon_0} \underbrace{\left[\hat{x} \times \left(y \hat{z} - z \hat{y} \right) \right]}_{\left[-y \hat{y} - z \hat{z} \right]}$

Energy flows in into the capacitor.

$$T^{00} = \frac{\varepsilon_0}{2}\vec{E}^2 + \frac{1}{2\mu_0}\vec{B}^2 = \frac{Q^2}{2\varepsilon_0(\pi R^2)^2} + \frac{\mu_0 \dot{Q}^2 r^2}{2(2\pi R^2)^2}$$

$$T^{01} = \frac{S_y}{c} = -\frac{Q\dot{Q}\,y}{2c\pi^2 R^4 \varepsilon_0}$$

$$T^{02} = \frac{S_z}{c} = -\frac{Q\dot{Q}\,z}{2\pi^2 c\,\varepsilon_0 R^4}$$

$$T^{03} = 0$$

$(i,j) \in 1,2,3$ $$T^{ij} = -\sigma^{ij} = \frac{1}{2\mu_0}\delta^{ij}\left(\frac{\vec{E}^2}{c^2} + \vec{B}^2\right) - \frac{1}{\mu_0}\left(\frac{E_i E_j}{c^2} + B_i B_j\right)$$

$$T^{xx} = -\frac{\varepsilon_0}{2} E_x^2 + \frac{1}{2\mu_0}\vec{B}^2 = -\frac{Q^2}{2\varepsilon_0 \pi^2 R^4} + \frac{\mu_0 \dot{Q}^2 r^2}{2(2\pi R^2)^2}$$

$$T^{yy} = \frac{\varepsilon_0}{2}\vec{E}^2 + \frac{1}{2\mu_0}\vec{B}^2 - \frac{1}{\mu_0}B_y^2 =$$

$$= \frac{Q^2}{2\varepsilon_0(\pi R^2)^2} + \frac{\mu_0 \dot{Q}^2}{2(2\pi R^2)^2}\underbrace{\left(r^2 - 2z^2\right)}_{y^2+z^2-2z^2 = y^2 - z^2.}$$

$$T^{zz} = \frac{\varepsilon_0}{2}\vec{E}^2 + \frac{1}{2\mu_0}\vec{B}^2 - \frac{1}{\mu_0}B_z^2 =$$

$$= \frac{Q^2}{2\varepsilon_0(\pi R^2)^2} + \frac{\mu_0 \dot{Q}^2}{2(2\pi R^2)^2}\left(z^2 - y^2\right)$$

$$T^{xy} = T^{xz} = 0$$

$$T^{yz} = -\frac{1}{\mu_0}B_y B_z = +\frac{\mu_0 \dot{Q}^2 yz}{(2\pi R^2)^2}$$

b) $$\mathcal{E} = \int T^{00}\, dV = \int_0^d dx \int_0^R dr\, r \int_0^{2\pi} d\varphi \; \frac{1}{(\pi R^2)^2}\left(\frac{Q^2}{2\varepsilon_0} + \frac{\mu_0 \dot{Q}^2}{8} r^2\right)$$

$$= \frac{2\pi d}{2\pi^2 R^4} \int_0^R dr\, r \left(\frac{Q^2}{\varepsilon_0} + \frac{\mu_0 \dot{Q}^2}{4} r^2\right)$$

Since $\dot{Q} = \text{const}$, $\frac{d\mathcal{E}}{dt} = \frac{d}{\varepsilon_0 \pi R^4} \int_0^R dr\, r \left(2 Q \dot{Q}\right) =$

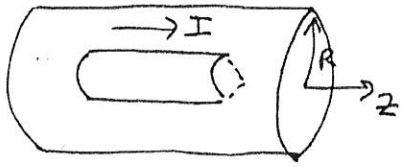
$$= \frac{d}{\pi \varepsilon_0 R^4}\, Q \dot{Q} R^2 = \frac{Q \dot{Q}\, d}{\varepsilon_0 \pi R^2}.$$

The meaning: potential between plates $V = Ed = \frac{Qd}{\varepsilon_0 \pi R^2}$

Current flowing $= \dot{Q}$

$\Rightarrow$ work done by outside source $V\dot{Q} = \frac{Q\dot{Q}d}{\varepsilon_0 \pi R^2}$.

P3



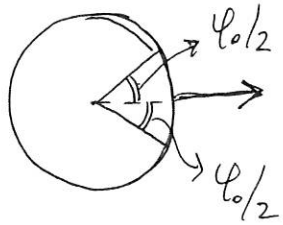
$$\vec{B} = \begin{cases} \dfrac{\mu_0 I}{2\pi r}\,\hat{\varphi} & \text{for } r > R \\ 0 & r < R \end{cases}$$

$$\sigma_{ij} = \frac{1}{\mu_0}\left(B_i B_j - \frac{1}{2}\delta_{ij}\vec{B}^2\right) = \begin{cases} -\dfrac{1}{2\mu_0}\left(\dfrac{\mu_0 I}{2\pi r}\right)^2 & \text{for } i=j=\hat{\varphi} \\ -\dfrac{1}{2\mu_0}\left(\dfrac{\mu_0 I}{2\pi r}\right)^2 & \text{for } i=j=\hat{\rho} \\ 0 & \text{otherwise} \end{cases}$$

Force acting on a segment in question is in radial direction (by symmetry)

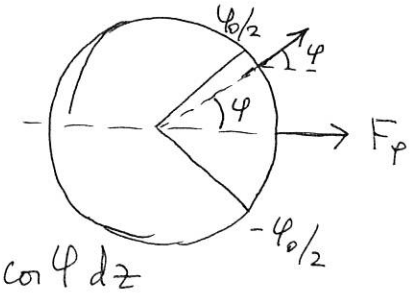
$$dF_\rho = \sigma_{\rho j}\, da_j = \underset{\substack{\parallel \\ 0}}{\sigma_{\rho\varphi}}\, da_\varphi + \sigma_{\rho\rho}\, da_\rho$$

For segment of angular width φ_0



resultant force F_ρ is along <u>the symmetry line of the segment</u>

$$F_\rho = \int \sigma_{\rho\rho}\, da_\rho \cos\varphi$$



Factor of <u>$\cos\varphi$</u> is need in order to calculate the component along the symmetry line.

$$= \int_{-\varphi_0/2}^{\varphi_0/2} \left[-\frac{1}{2\mu_0}\left(\frac{\mu_0 I}{2\pi R}\right)^2\right] \underbrace{R\,d\varphi \cos\varphi\, dz}_{da_\rho \cos\varphi}$$

$$= -\frac{1}{2\mu_0}\left(\frac{\mu_0 I}{2\pi R}\right)^2 RL \sin\varphi\Big|_{-\varphi_0/2}^{\varphi_0/2} = -\frac{\mu_0 I^2 L}{4\pi^2 R}\sin\left(\frac{\varphi_0}{2}\right).$$

Note that F_ρ points <u>inside</u>